Metadata

Title

Iconic duration effect in interpretation of sentences with and without descriptive attributes - determining the cut-off point of the effect

Description

An exploratory pilot study has shown that adding descriptive attributes to the noun phrases that refer to the participants of an event increases the perceived duration of the event in a task where the participants are explicitly asked to estimate the duration of the event described by the sentence by giving a number and a time unit. The effect seemed to be dependent on the inherent duration of the event. The effect was found in sentences that describe events that last up to one minute. No effect was found for events that last longer than one minute. However, the events were distributed unevenly within the bins (1-5 sec, 5-60 sec, 1-60 min, above 1 hour). In the 5-60 sec bin, most baseline durations were in the range of 5-20 seconds. The present experiment zooms in on that region in order to find the exact location of the cut-off point, after which the duration effect disappears. On the basis of baseline durations measured in a pretest, events (i.e. sentences that describe events) were put in bins of 0-10 sec, 10-20 sec, and 20-60 sec.

Contributors

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Subjects

Social and Behavioral Sciences Linguistics

Tags

duration iconicity narrative pragmatics semantics time

Study Information

Hypotheses

H1: If event duration is represented iconically by sentence length, then events described by longer sentences (with added descriptive attributes) are perceived to have longer duration

H2: The duration effect is bigger for the shorter events, and smaller or absent for events

? Help

Study Information

Hypotheses

H1: If event duration is represented iconically by sentence length, then events described by longer sentences (with added descriptive attributes) are perceived to have longer duration

H2: The duration effect is bigger for the shorter events, and smaller or absent for longer events

Design Plan

Study type

Experiment - A researcher randomly assigns treatments to study subjects, this includes field or lab experiments. This is also known as an intervention experiment and includes randomized controlled trials.

Blinding

For studies that involve human subjects, they will not know the treatment group to which they have been assigned.

Is there any additional blinding in this study?

No data

Study design

2x3 latin square design with two independent variables:
sentence length, with two levels: short (6-10 words), long (13-18 words)
inherent event duration, with three levels: 0-10 seconds, 10-20 seconds, 20-60 seconds
The stimuli are grouped into the three bins by the geometric means obtained in a pretest study. The actual geometric means can also be used directly as a continuous variable in the analysis.

No files selected

Randomization

The trials will be presented to the participant in a randomized order, using simple randomization across all experimental items and fillers.

Participants will be assigned randomly to two latin square lists, using balanced randomization.

Help

Sampling Plan

Existing Data

randomization.

Sampling Plan

Existing Data

Registration prior to creation of data

Explanation of existing data

No data

Data collection procedures

Participants will be recruited via Prolific (https://www.prolific.com/) and receive X GBP for their participation estimated to take X minutes. The following prescreeners will be applied:

Current Country of Residence: United Kingdom or Ireland
Country of Birth: United Kingdom or Ireland
First Language: English
Earliest Language in Life: English
Were you raised monolingual? I was raised with my native language only
Language related disorders: none
Literacy Difficulties: No
Number of previous submissions: 5–10000
Exclude participants from other studies:
- Experiment 1: Duration estimate single event (the exploratory pilot study)
- Experiment 2 Pretest: Iconic duration cut off point (the pretest)

Data will be collected online through the Gorilla Platform (https://gorilla.sc/). Participants will only be able to access the experiment if they use a desktop/laptop computer or a tablet.

No files selected

Sample size

Target sample: 60 participants, 30 per Latin square list.

Sample size rationale

No data

Stopping rule

No data

Variables

Help

Manipulated variables

sentence length, with 2 levels: short (6-10 words), long (13-18 words)

No data

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Manipulated variables

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No files selected

Measured variables

Participants will be asked to estimate the duration of the event described by the sentence by entering a number and choosing a time unit second(s) or minute(s).

No files selected

Indices

All provided values will be converted to seconds, i.e. multiplied by 60 if "minute(s)" is chosen as a time unit, or left unchanged otherwise.

No files selected

Analysis Plan

Statistical models

We will use a linear mixed-effects model with log-transformed duration estimates as the dependent variable. The model will include sentence length (short vs. long), event duration (three levels: 1-10 sec, 10-20 sec, 20-60 sec), and their interaction as fixed effects, as well as random effects of participant and item.

To test H1: a significant main effect of sentence length where long sentences are associated with longer durations would corroborate H1.

To test H2:
Our primary research question concerns whether the effect of sentence length differs across event duration levels—specifically, at which duration range this effect diminishes or disappears. To address this, we will conduct simple effects analyses (pairwise comparisons) testing the sentence length effect within each level of event duration separately, regardless of whether the omnibus interaction term reaches statistical significance.
We will apply Holm-Bonferroni correction to control for multiple comparisons across the three simple effects tests.

No files selected

Help

Transformations

No files selected

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No files selected

Transformations

The recorded event durations will be log transformed.

Inference criteria

We will use the standard 0.05 criteria for determining if the mixed effects models and the pairwise tests suggest that the results are significantly different from those expected if the null hypothesis were correct.

Data exclusion

No data

Missing data

No data

Exploratory analysis

No data

Help

Other

Other